Sarthak Kumar Mishra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.56
Chinmaya Vidyalaya (E.M.), Rourkela	Class XII (ISC)	2024	96.20%
Chinmaya Vidyalaya (E.M.), Rourkela	Class X (ICSE)	2022	98.60%

WORK EXPERIENCE

Research Intern - IIT Hyderabad - Prof. Shubhadeep Bhattacharjee

- Verified existing simulation of 4-bit Multiplier using stateful RRAM logic in Cadence Virtuoso using SKILL scripts
- Proposed an area-optimized b-bit multiplier and formulated area and latency figures using Python
- Identified redundancy in previous architecture and showed improvement in area and latency
- Installed AlmaLinux for Cadence Server and assisted in backup of previous machine
- Assumed ownership of an ongoing RRAM multiplier project following the graduation of the previous MTech researcher

PROJECTS

Pipelined RV32IZcaZmmul RISC-V Core - *GitHub*

- Designed a 5-stage pipelined RV32I RISC-V processor in SystemVerilog with GTKWave for debugging
- Implemented hazard detection, data forwarding, stalling, and pipeline flushing
- Developed a C++ testbench using Verilator to simulate, verify, and automate execution of ISA test programs
- Validated the implementation against the riscv-tests suite, passing 100% of the tests
- Integrated the Embench-IoT benchmark suite, achieving 1.17 CPI on the crc32 benchmark
- Implemented the Zca compressed instruction extension, reducing crc32 benchmark code size by 37.3%
- Designed a 32-bit multi-cycle Booth Multiplier module to multiply signed and unsigned numbers
- Integrated the multiplier to implement the Zmmul extension, reducing cycle count by 70.7% for the edn benchmark

FPGA Mazesolver - *E-Yantra Robotics Competition, IIT Bombay (2025-26)*

- Collaborated in a four-member team to design, build, and assemble a Cyclone IV DE0-Nano FPGA-based robot
- Implemented a modified Trémaux algorithm for exploring perfect mazes in Verilog using fewer than 800 LUTs
- Achieved maze completion in 184 steps while exploring all 9 dead ends in the competition-provided maze
- Deployed the robot and navigated a physical maze in 240 seconds without human intervention during the run
- Simulated the DHT sensor protocol, ultrasonic sensor interface, UART and a single-cycle RV32I core in Verilog

Drift-Diffusion-based MOSCAP Simulator - *GitHub*

- Developed transient drift-diffusion-based 1D MOSCAP simulator with variable generation rate in C
- Derived and implemented Scharfetter-Gummel discretization and Crank-Nicolson for the carrier continuity equation
- Implemented a 1D Poisson solver and integrated it in the simulation loop using Gummel iterations
- Tuned parameters for numerical stability and convergence
- Parallelized computation for each bias using OpenMP, achieving 10× speedup
- Plotted and compared the C-V curves in Gnuplot, capturing the effect of different generation rates and time evolution

STM32-based Macropad - *GitHub*

- Collaborated with 5 people and designed a Macropad based on the STM32F103C8T6 microcontroller
- Developed the firmware for SSD1306 display using libopencm3
- Debugged firmware and hardware using OpenOCD, GDB, and ST-LINK, validating USB functionality
- Designed a custom enclosure and mechanical components in Onshape and fabricated it using 3D printing
- Developed a GUI in C using microui for generating key mappings and flashing over USB.
- Managed PCB fabrication logistics, including quotation, component shipping, and manufacturer communication

Custom Shell - *CodeCrafters* - *GitHub*

- Implemented a Unix-like shell supporting built-in commands and execution of external programs through PATH resolution
- Developed a command tokenizer handling quotes, escape sequences, I/O redirections, and pipe operators
- Added support for output/error stream redirection and multi-stage pipelines with integration of built-in commands

SKILLS

Languages: C, C++, Python, SystemVerilog, Verilog

EDA & Simulation: LTspice, NGspice, Cadence Virtuoso

FPGA Toolchain: Verilator, Icarus Verilog, GTKWave, Quartus Prime, OSS CAD Suite (Yosys, nextpnr)

Embedded: OpenOCD, GDB, libopencm3

PCB Design: KiCad, Altium Designer

POSITIONS OF RESPONSIBILITY

Elektronica - Domain Head

Elektronica - Core Member

- Conducted sessions on Computer Architecture and RTOS
- Designed and evaluated the problem statement for the Spring Bootcamp
- Represented the club in E-Yantra Robotics Competition (eYRC), IIT Bombay 2025-26 in the Mazesolver Bot theme